

A conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

18 September 2026

Summary

We adapted the DINO-Fusion diffusion pipeline to generate the post-spin-up temperature and salinity state of the Veros ACC channel configuration directly from the two coefficients of its eddy-kinetic-energy closure, c_k and c_ϵ . The pipeline runs end to end on Jean Zay in under one A100 hour. In this first run, with DINO-Fusion’s per-level normalisation, the conditional model reproduces the mean training state rather than the parameter dependence: on ten held-out parameter pairs the eight-sample ensemble mean has an RMSE of 0.154 K against the true 20-year mean state, the same as the mean-state baseline (0.155 K), while the average of the four neighbouring runs on the grid reaches 0.020 K. Two harder hold-out splits, a band of three interior rows and the top row, show that the model’s error is nearly independent of the split while the interpolation baselines degrade with the gap, and reveal a physical miss: the model generates temperature inversions at the rate of the weakly eddying runs whatever the regime.

1 Data

One hundred Veros ACC runs on a complete 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$ and $c_\epsilon \in [0.0875, 5.6]$, 100 years each with 30-day output on a $30 \times 42 \times 15$ interior grid (zonal, meridional, vertical). We keep temperature and salinity for the last 20 years of every run (241 snapshots per run). Three properties of the ensemble shape the problem:

- Salinity is exactly 35 psu in every water cell of every run; it is carried through the pipeline as a second field but contains no information.
- The end states are nearly deterministic: the grid-point temporal standard deviation inside the window is 0.01 K above 1400 m, against 0.05 K between neighbouring runs on the grid and up to 1.5 K between extremes. The deep layers are the exception: the bottom level still drifts by 0.45 K on average over the last 20 years, as much as the spread between regimes, so the target there is a slowly moving state rather than an equilibrium.
- The two parameters act almost as one: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean temperature, and the low- c_k/c_ϵ part of the grid saturates to one state. The response lives between 600 and 800 m and near the bottom, not at the restored surface.

Ten interior grid points were held out at random before any statistic was computed; the remaining 90 runs (21 690 snapshots) form the training set.

2 Method

Temperature and salinity are stacked into 30 channels, normalised per vertical level as in DINO-Fusion, $x' = (x - \mu_z) / (3\sigma_z)$ with statistics from the training runs only and a floor on σ_z that keeps the constant

salinity at exactly zero, set to zero on the fixed land mask, and zero-padded from 42×30 to 48×32 so the UNet can halve the grid four times. With this scaling 99.4 % of the training temperature values lie inside the sampler’s clipping range $[-1, 1]$; the exceptions are the cold southern boundary between 650 and 1400 m and the two bottom levels, whose extremes are clipped at sampling. The denoiser is a four-level UNet2DModel (64, 64, 128, 128 channels, 6.9 M parameters). The standardised $(\log c_k, \log c_\epsilon)$ pair passes through a small MLP whose output is added to the timestep embedding. Training is the standard DDPM ϵ -prediction objective with 1000 steps, a squared-cosine schedule, sample clipping, AdamW at 10^{-4} with cosine decay, batch 32, 20 000 steps and an EMA of the weights. Sampling runs the 1000-step DDPM loop with the EMA weights and re-imposes zeros on land and padding after every step.

3 Evaluation

For each held-out run we draw eight samples at its (c_k, c_ϵ) and compare, on water cells, with the true time-mean state of the last 20 years. Baselines: the mean training state, the nearest training run in standardised log-parameter space, and the average of the axis-neighbour training runs on the grid. We also generate four samples for every one of the 100 grid points to compare the map of domain-mean temperature with the truth.

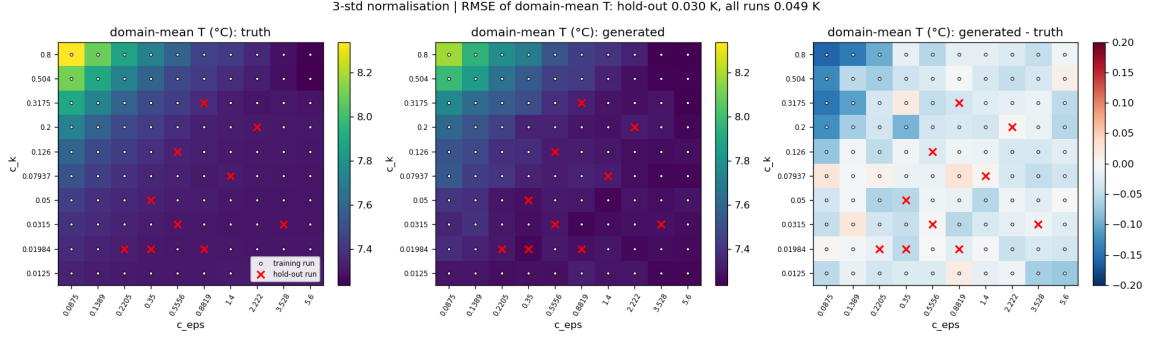
	per-level normalisation
diffusion, ensemble mean of 8: RMSE (K)	0.154 ± 0.018
diffusion, single sample: RMSE (K)	0.188 ± 0.015
neighbour average: RMSE (K)	0.020 ± 0.011
nearest training run: RMSE (K)	0.037 ± 0.027
mean training state: RMSE (K)	0.155 ± 0.030
diffusion: domain-mean bias (K)	-0.028
ensemble spread / true within-window spread (K)	$0.105 / 0.031$
interfaces with T decreasing upward, generated / truth	$9.5 \% / 7.7 \%$
max salinity error (psu)	0.010
domain-mean T map, RMSE on hold-out / all runs (K)	$0.030 / 0.049$

Table 1: Hold-out metrics, mean $\pm$ standard deviation over the ten held-out runs, water cells only.

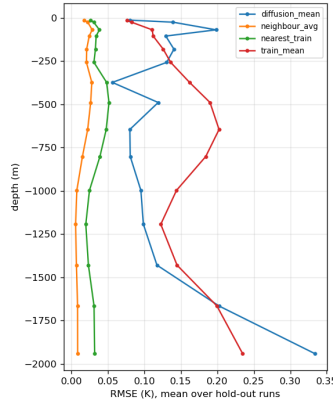
4 Findings

1. **The pipeline works end to end.** Extraction, training (22 minutes on one A100), conditional generation and evaluation ran as one SLURM dependency chain in under one GPU hour, and every component was checked on synthetic and real data before the run.
2. **Conditioning is not yet effective.** The ensemble-mean RMSE equals the mean-state baseline: the model reproduces the structure shared by all runs and adds noise. The generated map of domain-mean temperature over the grid (Figure 1a) shows only a faint response in the warm high- c_k , low- c_ϵ corner. The mechanism is quantitative: under per-level scaling the run-to-run signal is 0.7 % of the normalised range at the surface, 9 % at 650 m and 25 % at the bottom, and the ϵ -prediction loss weights every cell equally, so training is dominated by the shared structure. Consistently, the model beats the mean-state baseline only between about 400 and 1400 m (Figure 1b); it is worse than it in the top 250 m, where the signal is below one percent of the range, and at the two bottom levels, where the training distribution is broadened by the residual drift and its extremes are clipped.
3. **Spread.** Single samples are 0.188 K from the truth and the ensemble spread is three times the true within-window spread; the samples in Figure 4 look like plausible states but differ from the truth at mid-depth and below.

4. **Plumbing checks pass.** Salinity returns within 0.01 psu of 35 without any constraint, land is exact, and the fraction of temperature inversions (9.5 %) is close to the truth’s (7.7 %).



(a) Domain-mean temperature over the parameter grid: truth, generated, difference. White dots: training runs; red crosses: held-out runs.



(b) RMSE against the truth by depth, mean over the held-out runs: diffusion ensemble mean and baselines.

Figure 1: Scattered split (10 interior held-out runs), 32 samples per run.

5 Hold-out splits

The scattered split gives every held-out run a training neighbour one grid step away on all four sides, the easiest case for an interpolation baseline. Two harder splits were run with identical settings and 32 samples per held-out run: a contiguous band of three interior rows ($c_k = 0.126, 0.2$ and 0.3175 , 30 runs, so the nearest training rows are a factor 6.3 apart in c_k) and the top row ($c_k = 0.8$, 10 runs, pure extrapolation). Each split has its own training set and normalisation statistics. An additional baseline interpolates linearly in $\log c_k$ between the nearest training rows below and above in the same c_ϵ column; for the top row it reduces to copying the row below. The scattered split was re-evaluated with 32 samples for consistency (0.150 K against 0.154 K with 8).

	scattered 10 runs	band of three rows 30 runs	top row 10 runs
diffusion, ensemble mean of 32: RMSE (K)	0.150	0.129	0.193
diffusion, single sample: RMSE (K)	0.186	0.175	0.232
interpolation in $\log c_k$: RMSE (K)	—	0.083	0.170
nearest training run: RMSE (K)	0.037	0.116	0.170
mean training state: RMSE (K)	0.155	0.197	0.470
diffusion: domain-mean bias (K)	−0.024	−0.023	−0.083
ensemble spread / true spread (K)	0.110 / 0.031	0.113 / 0.031	0.114 / 0.032
T inversions (interfaces), generated / truth	9.6 % / 7.7 %	11.0 % / 1.3 %	12.0 % / 0.1 %

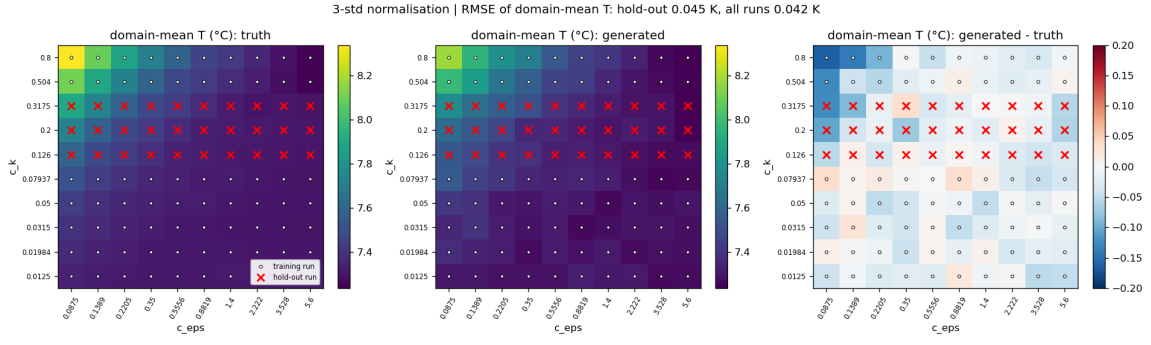
Table 2: The three hold-out splits, mean over the held-out runs, water cells only, 32 samples per run.

Three points follow. First, the model’s error is nearly independent of the split (0.13 to 0.19 K) while the baselines’ error grows with the gap they must bridge (0.02 to 0.17 K); the model therefore catches up as the split gets harder, beats the nearest training run in the two upper rows of the band and over the trending part of the top row (c_ϵ from 0.22 to 0.88), but never beats interpolation in $\log c_k$ over an interior gap. Within the band it is the best method between 650 and 800 m (0.08 K against 0.11 K for interpolation); its floor comes from the top 250 m and the two bottom levels, as before. Second, extrapolation is bounded before the model is: under the top split’s own training statistics, 17 % of the true bottom-level values of the held-out row lie beyond the sampler’s clip range and cannot be generated, so the generated corner is 0.2 K too cold. Third, the inversion rows of the table show a physical miss that grows with c_k : the model generates 10 to 12 % of interfaces with temperature decreasing upward whatever the regime, the rate of the weakly eddying rows that dominate the grid, whereas in the truth the strongly eddying states are stratified almost everywhere. The model has not learned that regime dependence, and this is precisely the kind of constraint DINO-Fusion imposes at sampling time. Figures 2 and 3 show the parameter map and the depth profile for the two splits; Figures 4, 5 and 6 show the visual check for one held-out condition of each split.

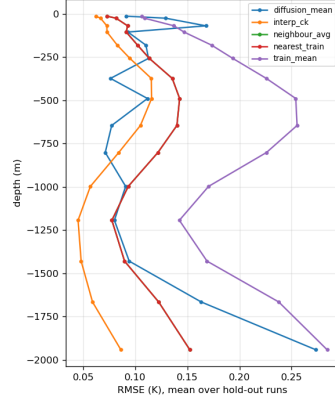
6 Next steps

Impose stratification at sampling time, DINO-Fusion’s isotonic projection applied to the temperature profile since salinity is constant, which targets the inversion miss directly. Strengthen the conditioning within this normalisation: classifier-free guidance, a longer schedule or a larger conditioning path, and the year as a third condition so the model knows where in the drift window a state sits. Evaluate against snapshots as well as the time mean, since the training targets drift by up to 0.45 K at the bottom inside the window. Raise the clip range or the scale factor where extrapolation matters, since the top-row test was bounded by clipping. The longer-term target is the same pipeline on a global Veros configuration, where active salinity, seasonal forcing and multi-century spin-up make the generative formulation and the density constraints of DINO-Fusion relevant.

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`, commit `f912e66` for the scattered-split run and `5991188` for the two other splits. Extraction takes 2.5 minutes on 8 CPU cores, training 22 minutes on one A100, generation and evaluation 3.5 minutes; the whole dependency chain ran on the development queue in under one GPU hour.

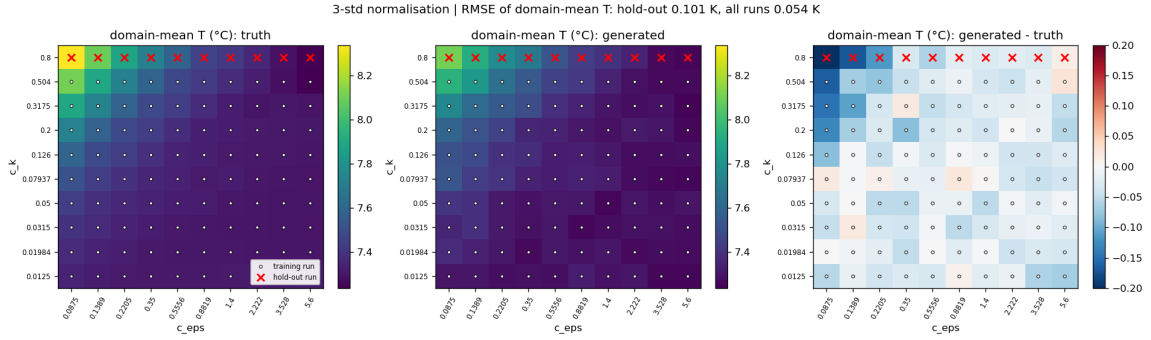


(a) Domain-mean temperature over the parameter grid.

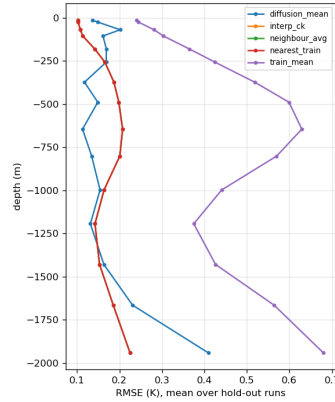


(b) RMSE by depth, mean over the 30 held-out runs; the interpolation baseline is the orange curve.

Figure 2: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.



(a) Domain-mean temperature over the parameter grid.



(b) RMSE by depth, mean over the 10 held-out runs; all three baselines coincide (the row below).

Figure 3: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.

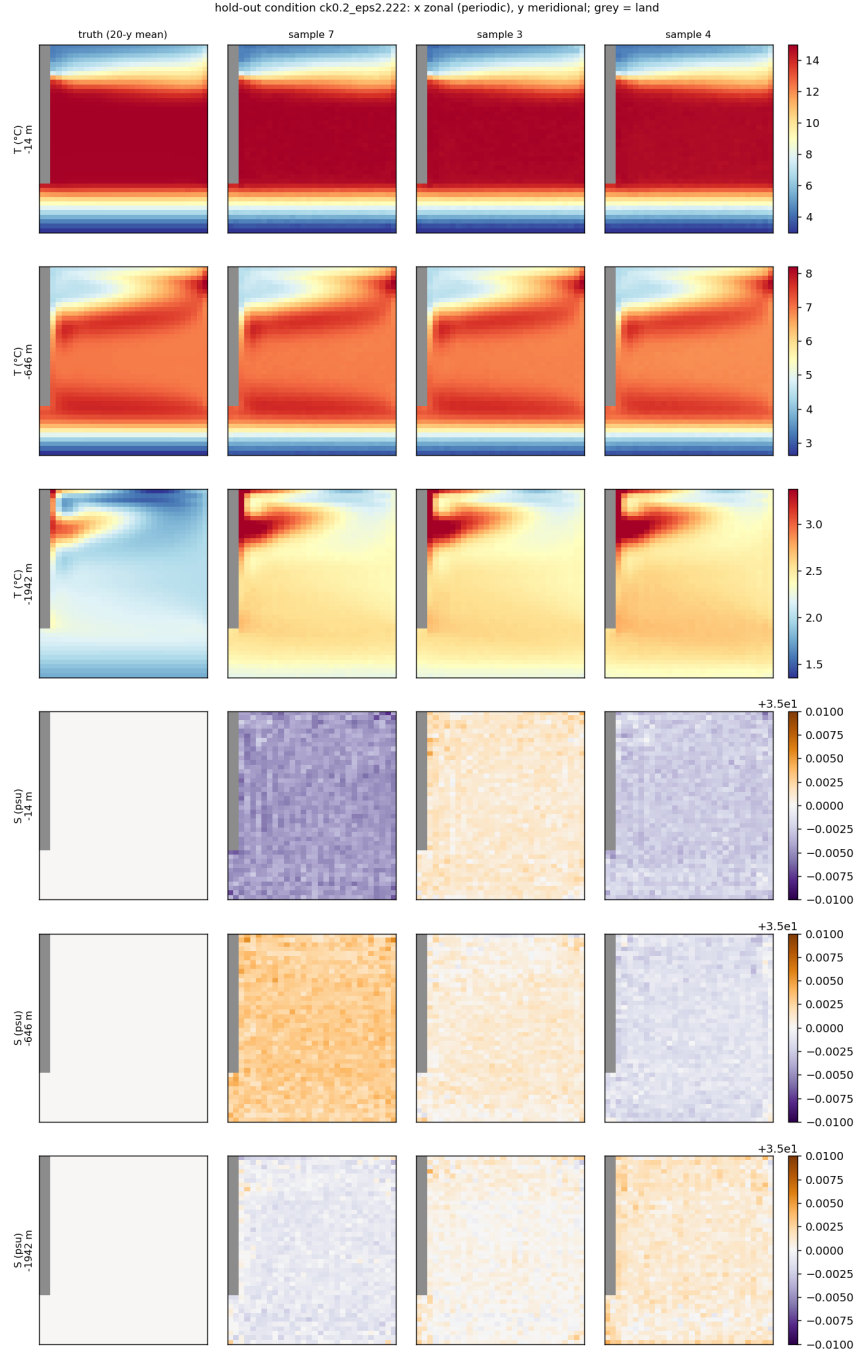


Figure 4: Scattered split, visual check for one held-out condition ($c_k = 0.2$, $c_\epsilon = 2.222$): the true 20-year mean state (left column) and three random samples at the surface, at about 650 m and at the bottom, for temperature (top three rows) and salinity (bottom three rows). Salinity is 35 psu everywhere in the truth; the salinity panels show the residual noise of the generator on a colour scale of a few hundredths of a psu. Grey is land.

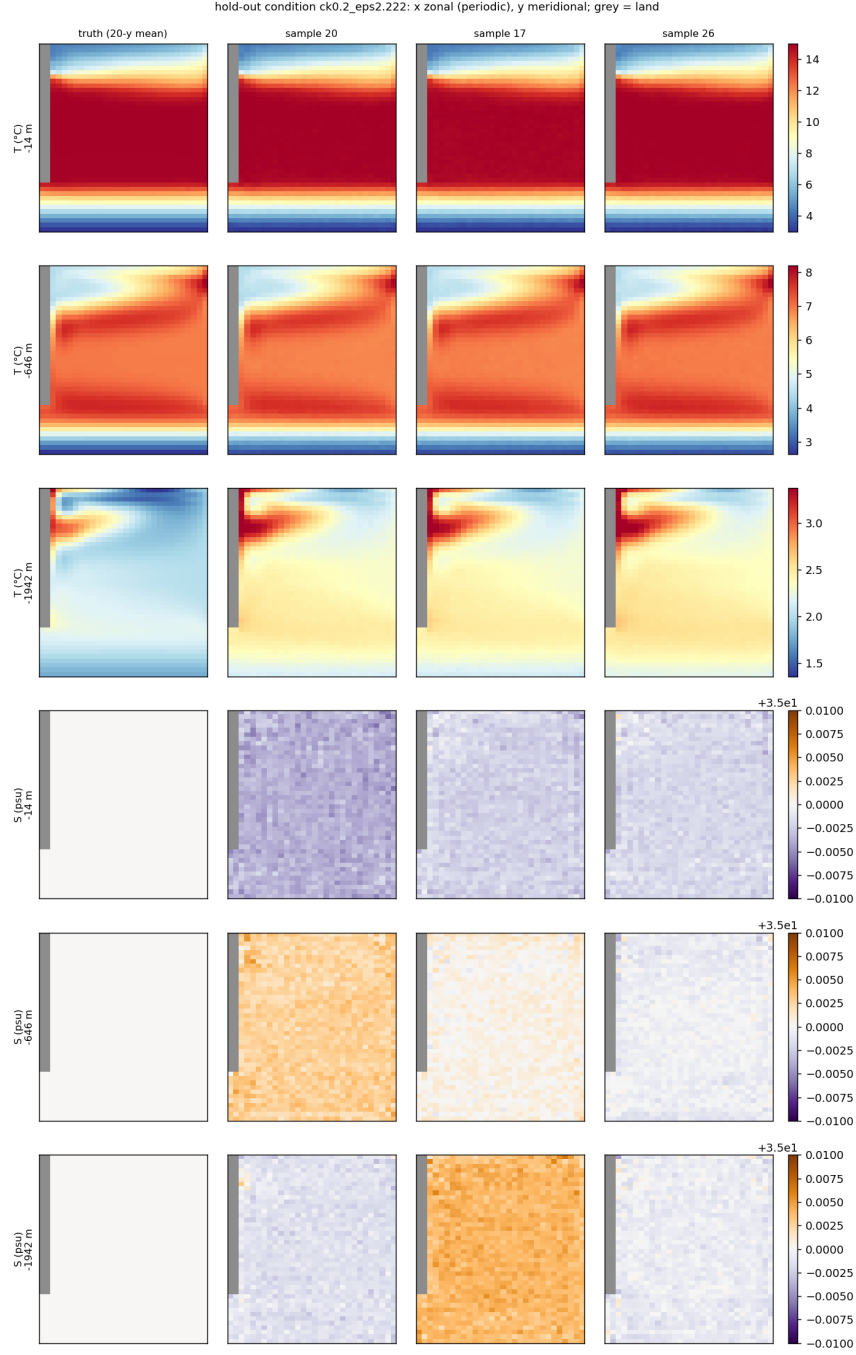


Figure 5: Band-of-three split, same visual check for the same condition ($c_k = 0.2$, $c_\epsilon = 2.222$), which now sits in the middle held-out row: true 20-year mean state and three random samples at the surface, about 650 m and the bottom, temperature (top) and salinity (bottom). Grey is land.

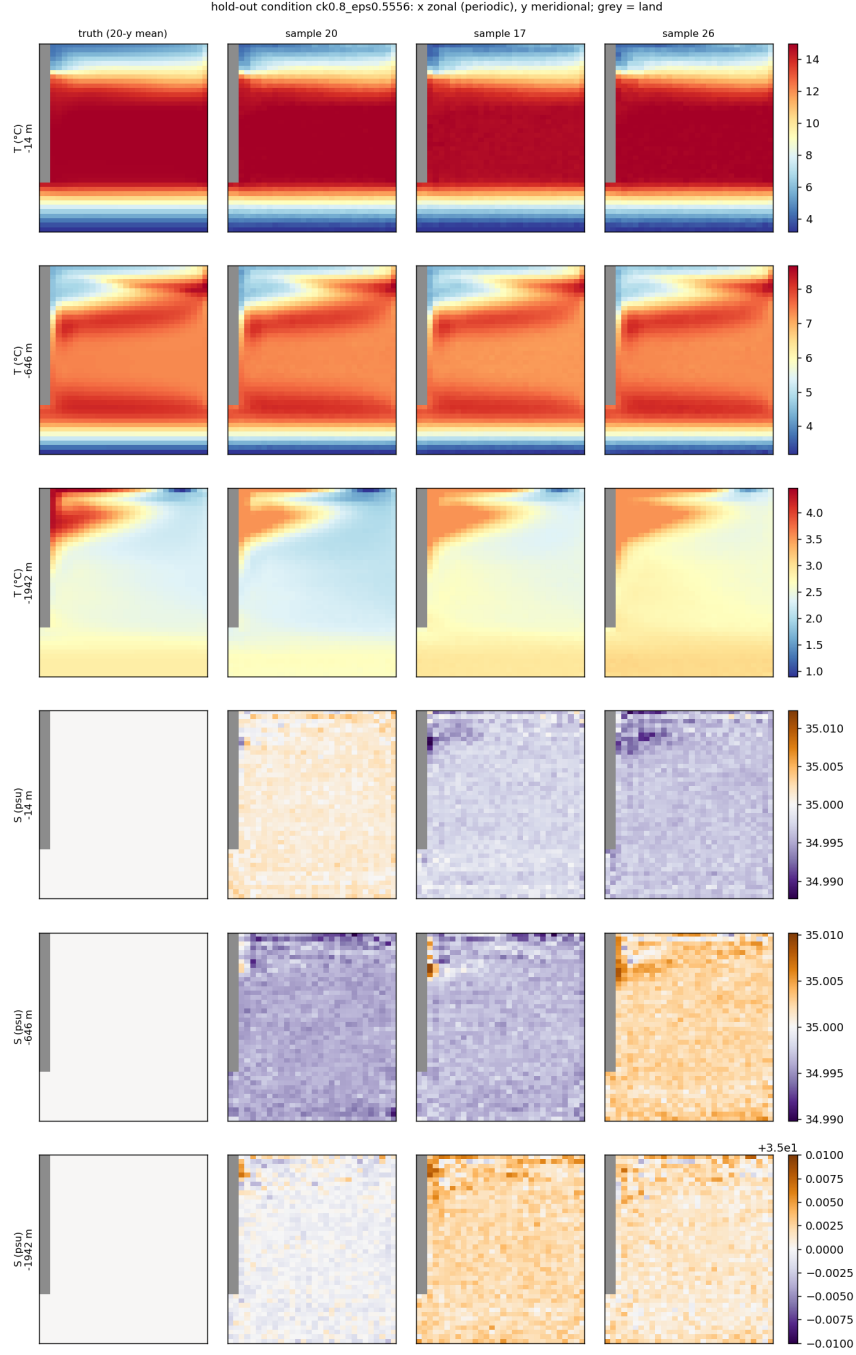


Figure 6: Top-row split, visual check for the held-out condition $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid): true 20-year mean state and three random samples at the surface, about 650 m and the bottom, temperature (top) and salinity (bottom). Grey is land.